

HiveForce Labs

THREAT ADVISORY



ATTACK REPORT

Silver Fox Slips ValleyRAT Into India Through Fake Tax Notices

Date of Publication

December 31, 2025

Admiralty Code

A1

TA Number

TA2025394

Summary

Attack Commenced: December 2025

Threat Actor: Silver Fox (alias Void Arachne)

Malware: Donut loader, Valley RAT

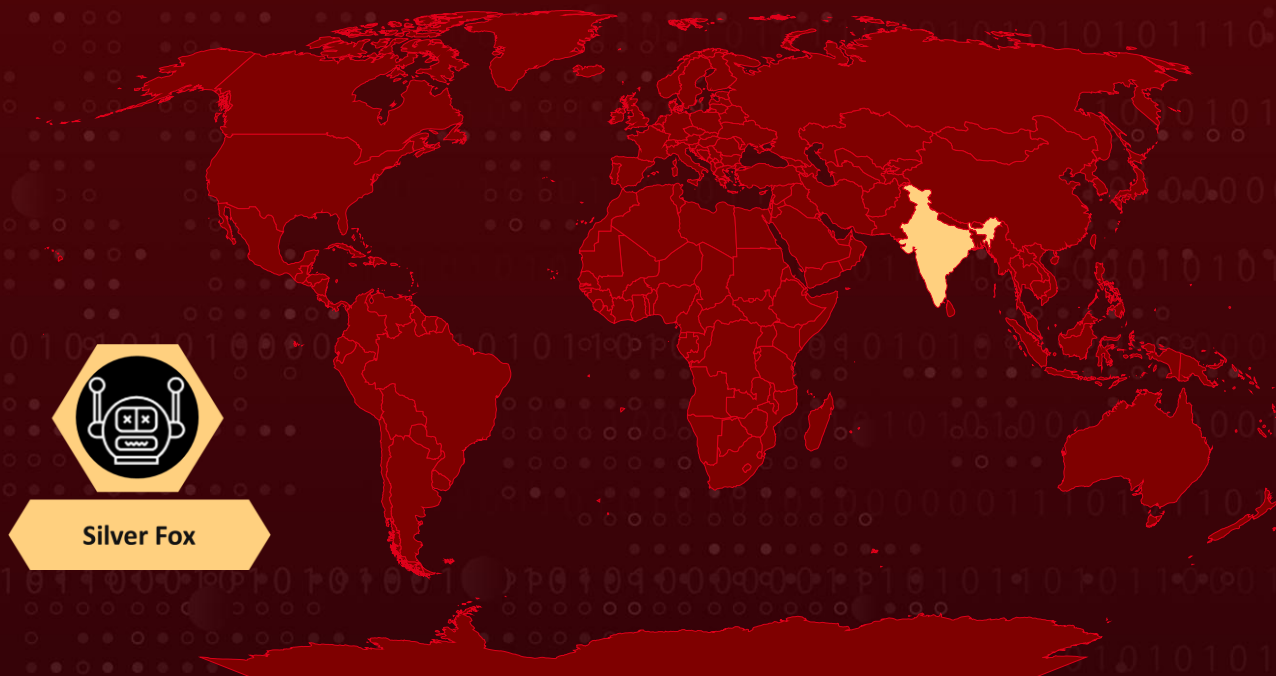
Targeted Region: India

Targeted Industries: Enterprise, Finance, Medical, Technology

Affected Platform: Windows

Attack: Silver Fox, a Chinese-aligned advanced persistent threat group, has conducted a highly sophisticated phishing campaign against Indian organizations by impersonating official Income Tax Department communications to stealthily deploy the modular ValleyRAT malware. The malware leverages registry-based persistence, tiered command-and-control infrastructure with failover, and a plugin-driven architecture that enables credential theft, keylogging, and defense evasion, underscoring the group's high operational maturity and evolving cyber-espionage capabilities.

🔪 Targeted Regions



Silver Fox

Powered by Bing
© Australian Bureau of Statistics, GeoNames, Microsoft, Navinfo, Open Places, OpenStreetMap, TomTom, Zenrin

Attack Details

#1

Silver Fox, a Chinese-aligned advanced persistent threat group, executed a highly refined phishing operation against Indian organizations by masquerading as legitimate correspondence from the Income Tax Department. This activity marks the first confirmed instance of tax-themed phishing being attributed to Silver Fox.

#2

The operation unfolds as a carefully staged intrusion designed for stealth and longevity. It begins with convincing phishing emails carrying a malicious PDF crafted to resemble official tax communications. When opened, the document silently initiates the download of a rogue executable "tax affairs.exe" packaged as a Nullsoft Scriptable Install System installer and serving as the entry point for the attack.

#3

From there, the campaign escalates through a sophisticated infection chain. The installer exploits DLL hijacking within legitimate software, disables Windows Update to weaken host defenses, and deploys a Donut loader after conducting extensive anti-analysis and anti-sandbox checks. The malware then uses process hollowing to inject its final payload into a legitimate process, effectively concealing its presence.

#4

At the core of the intrusion is ValleyRAT, a modular remote access trojan engineered for persistence and adaptability. It maintains covert communication with external command-and-control servers built with tiered infrastructure and failover resilience. Its plugin-based design allows operators to selectively deploy capabilities such as keylogging, credential theft, and advanced defense evasion, ensuring sustained and flexible control over compromised systems.

Recommendations



Monitor registry abuse as a persistence mechanism: Implement continuous monitoring for executable REG_BINARY values and anomalous registry entries under non-standard paths such as HKCU\Console*, with scrutiny on keys written by user-level processes rather than trusted system components.



Instrument memory permission and injection anomalies: Alert on processes allocating PAGE_EXECUTE_READWRITE memory followed by remote or local thread creation, especially when these actions occur inside normally benign processes such as explorer.exe, indicating process hollowing or in-memory payload staging.



Hunt for signed binary abuse with local DLL hijacking: Correlate execution of signed executables launched from temporary or user-writable directories with the loading of unsigned local DLLs, particularly when followed by immediate thread creation or abnormal execution flow.



Endpoint and Network Defense Enhancements: Deploy endpoint detection and response tools tuned to identify DLL sideloading and process injection attempts. Monitor for suspicious persistence methods, including services, scheduled tasks, and registry entries. Inspect network traffic for command-and-control activity.



Potential MITRE ATT&CK TTPs

<u>TA0001</u> Initial Access	<u>TA0002</u> Execution	<u>TA0003</u> Persistence	<u>TA0005</u> Defense Evasion
<u>TA0007</u> Discovery	<u>TA0009</u> Collection	<u>TA0011</u> Command and Control	<u>TA0040</u> Impact
<u>T1566</u> Phishing	<u>T1566.001</u> Spearphishing Attachment	<u>T1204</u> User Execution	<u>T1204.002</u> Malicious File
<u>T1059</u> Command and Scripting Interpreter	<u>T1106</u> Native API	<u>T1129</u> Shared Modules	<u>T1620</u> Reflective Code Loading



<u>T1547</u> Boot or Logon Autostart Execution	<u>T1112</u> Modify Registry	<u>T1574</u> Hijack Execution Flow	<u>T1574.001</u> DLL
<u>T1218</u> System Binary Proxy Execution	<u>T1027</u> Obfuscated Files or Information	<u>T1497</u> Virtualization/Sandbo x Evasion	<u>T1562</u> Impair Defenses
<u>T1562.001</u> Disable or Modify Tools	<u>T1057</u> Process Discovery	<u>T1082</u> System Information Discovery	<u>T1071</u> Application Layer Protocol
<u>T1071.001</u> Web Protocols	<u>T1095</u> Non-Application Layer Protocol	<u>T1105</u> Ingress Tool Transfer	<u>T1573</u> Encrypted Channel
<u>T1008</u> Fallback Channels	<u>T1041</u> Exfiltration Over C2 Channel	<u>T1056</u> Input Capture	<u>T1056.001</u> Keylogging
<u>T1547.001</u> Registry Run Keys / Startup Folder	<u>T1489</u> Service Stop		

✂ Indicators of Compromise (IOCs)

TYPE	VALUE
SHA256	77ea62ff74a66f61a511eb6b6edac20be9822fa9cc1e7354a8cd6379c7b9d2d2, fa388a6cdd28ad5dd83acd674483828251f21cbefaa801e839ba39af24a6ac19, f74017b406e993bea5212615febe23198b09ecd73ab79411a9f6571ba1f94cfa, 068e49e734c2c7be4fb3f01a40bb8beb2d5f4677872fabbced7741245a7ea97c,
Domians	ggwk[.]cc, b[.]yuxuanow[.]top, itdd[.]club, xzghjec[.]com, gov-a[.]work, gov-a[.]fit, gvo-b[.]club, gov-c[.]club, gov-a[.]club, govk[.]club,

TYPE	VALUE
Domains	dingtalki[.]cn, hhiioo[.]cn, kkyui[.]club, hhimm[.]work, swjc2025bjkb[.]cn, 2025swmm[.]cn, hhiioo[.]work
IPv4	45[.]207[.]231[.]94, 103[.]20[.]195[.]147, 45[.]207[.]231[.]107, 8[.]217[.]9[.]165, 160[.]124[.]9[.]103, 47[.]239[.]225[.]43, 43[.]100[.]22[.]158, 43[.]100[.]123[.]207, 43[.]100[.]63[.]145

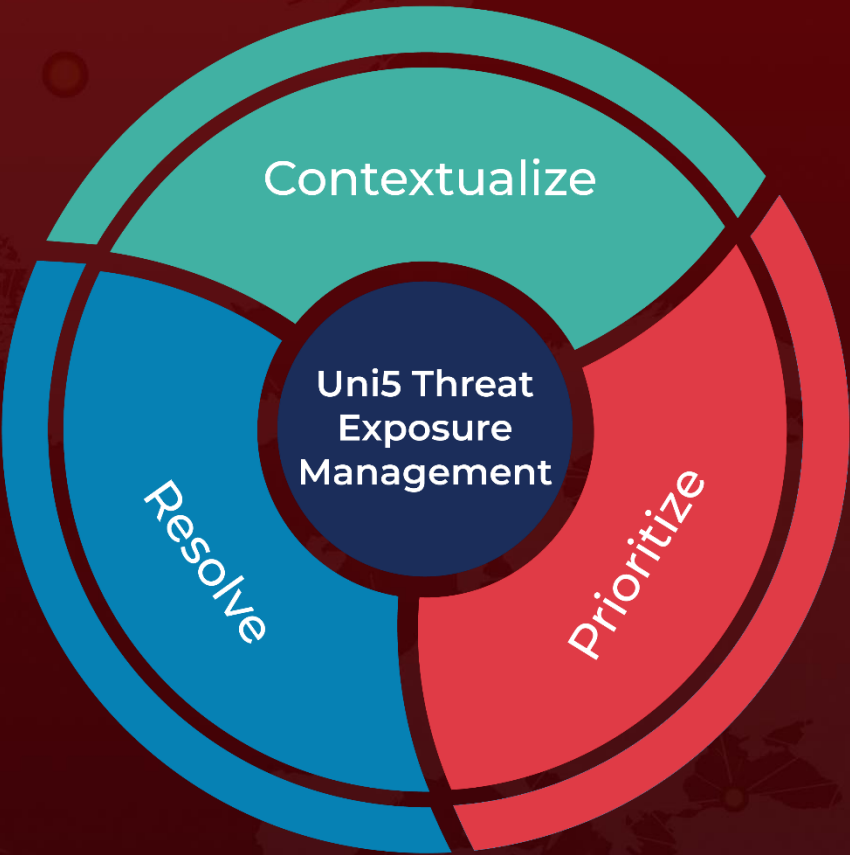
References

<https://www.cloudsek.com/blog/silver-fox-targeting-india-using-tax-themed-phishing-lures>

What Next?

At Hive Pro, it is our mission to detect the most likely threats to your organization and to help you prevent them from happening.

Book a free demo with HivePro Uni5: Threat Exposure Management Platform.



REPORT GENERATED ON

December 31, 2025 • 6:00 AM

© 2025 All Rights are Reserved by Hive Pro



More at www.hivepro.com